

Project 2: Adversarial robustness and probabilistic calibration

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This project investigates the trade-off between adversarial robustness and probabilistic calibration in deep neural networks. Using the CIFAR-10 dataset, we implement and compare four configurations across a Small CNN and a ResNet-18 under standard and FGSM-based adversarial training. We evaluate robustness against FGSM and PGD attacks across multiple perturbation strengths ($\epsilon \in \{0, 4/255, 8/255, 12/255, 16/255\}$), and measure calibration via Expected Calibration Error (ECE), Negative Log-Likelihood (NLL), and reliability diagrams. Our results confirm that adversarial training improves robustness against FGSM and PGD attacks, but introduces a robustness tax on clean accuracy and mild degradation in clean-data calibration. Standard models, while well-calibrated on clean data, exhibit catastrophic overconfidence under attack. Scaling to ResNet-18 amplifies both the benefits and costs of adversarial training without resolving the fundamental trade-off.

Index Terms—Adversarial Training, FGSM, PGD, Calibration, Expected Calibration Error, ResNet-18, CIFAR-10, Robustness-Calibration Trade-off.

I. INTRODUCTION

TRADITIONAL Deep neural networks are very vulnerable to adversarial attacks, which are malicious perturbations of the data designed to cause confident misclassifications. **Adversarial Training** is the most effective defense. However, these models tend to exhibit degraded probabilistic calibration, meaning that they are less reliable. This report investigates the trade-off between the robustness and the calibration on CIFAR-10 across two models: a small Convolutional Neural Network and a ResNet-18.

II. EXPERIMENTAL SETUP

A. Dataset and Pre-processing

All experiments use CIFAR-10, which includes 50,000 training and 10,000 test images divided into 10 classes, each of size 32×32 in RGB. Images are loaded with a single `ToTensor()` transformation, keeping pixel values in $[0, 1]$. No data augmentation is applied, ensuring calibration measurements are not affected by these augmentations. Batch size is 128 throughout.

B. Architectural Backbone

1) *Small CNN*: A custom four-layer convolutional network with ReLU activations is used. It includes 2×2 max-pooling

after each block, reducing spatial resolution from 32×32 to 2×2 . Two fully-connected layers ($256 \rightarrow 10$) produce the class logits. Total parameters: ≈ 2.4 M.

2) *ResNet-18*: A standard ResNet-18 is adapted for CIFAR-10 by replacing the initial 7×7 convolution with a 3×3 kernel of stride 1 and removing the early max-pooling layer to prevent excessive spatial downsampling on 32×32 inputs. The final fully-connected layer outputs 10 classes. Total parameters: ≈ 11.2 M.

C. Training Protocols

Both architectures are trained for 15 epochs using Adam with cross-entropy loss. Standard training uses $\text{lr} = 1e-3$. Adversarial training uses $\text{lr} = 1e-3$ for the Small CNN and $\text{lr} = 1e-4$ for ResNet-18. All other hyperparameters are constant to isolate the effect of the training objective.

D. Attacks

1) *FGSM (Fast Gradient Sign Method)*: Single-step attack: $x' = x + \epsilon \cdot \text{sign}(\nabla_x L(\theta, x, y))$.

2) *PGD-10 (Projected Gradient Descent)*: Iterative attack over 10 steps with step size $\alpha = \epsilon/4$ and random initialisation within the ϵ -ball. Clipping to $[0, 1]$ is applied after each step.

Adversarial training injects FGSM examples at each minibatch with $\epsilon = 8/255$. Evaluation sweeps $\epsilon \in \{0, 4/255, 8/255, 12/255, 16/255\}$.

E. Calibration Metrics

1) *Expected Calibration Error (ECE)*: Partitions predictions into 10 equally-spaced confidence bins and computes a weighted average of $|\text{accuracy} - \text{confidence}|$ per bin. The lower, the better.

2) *Negative Log-Likelihood (NLL)*: Cross-entropy of the softmax distribution over true labels. It penalizes a lot incorrect predictions made with high confidence. It also penalizes correct predictions made with high uncertainty, encouraging the model to be accurate and sure.

3) *Reliability Diagrams*: Visual comparison of mean confidence vs. true accuracy per bin. A perfectly calibrated model lies on the diagonal.

III. RESULTS

A. Part 1 — Small CNN

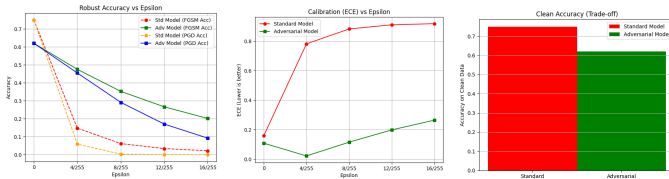
Metric	Standard	FGSM-AT	Diff.
Clean Accuracy (%)	74.96	62.00	−12.96
FGSM Robust Acc (%)	6.20	35.25	+29.05
PGD Robust Acc (%)	0.29	29.13	+28.84
Clean ECE	0.1591	0.1084	−0.0507
FGSM ECE	0.8835	0.1152	−0.7683
PGD ECE	0.9955	0.1796	−0.8159

TABLE I: Small CNN: Standard vs. FGSM-AT.

Standard training achieves strong clean accuracy but is essentially defenseless against adversarial attacks, with robust accuracy collapsing to single digits under PGD. FGSM-based Adversarial Training recovers substantial robustness: FGSM accuracy rises to $\sim 35\%$ and PGD accuracy to $\sim 29\%$, confirming that the defense generalises to the stronger iterative attack. The cost is a ~ 13 pp drop in clean accuracy.

Regarding calibration, the standard model is well-calibrated on clean data ($ECE \approx 0.16$), but under attack it becomes maximally overconfident, assigning high confidence to wrong predictions. The adversarially trained model shows a slightly lower clean ECE (~ 0.11) and higher NLL, indicating a bit overconfidence even on clean data. Reliability diagrams confirm this: on clean inputs the adversarial model's bars are slightly above the diagonal, which means there is underconfidence. While under FGSM or PGD attack both models' bars fall below the diagonal, leading to overconfidence, with the adversarial model being less extreme.

As ε increases, ECE grows quadratically for the standard model and more gradually for the adversarial model. The adversarial model shows a slight ECE decrease at $\varepsilon = 4/255$ before climbing, suggesting the training distribution ($\varepsilon = 8/255$) confers a calibration buffer at lower perturbation levels.



(a) Robust Acc. (b) Calibration (ECE) (c) Clean Trade-off

Fig. 1: Small CNN analysis.

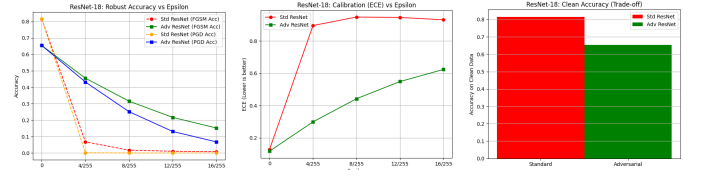
B. Part 2 — ResNet-18

Metric	Standard	FGSM-AT	Diff.
Clean Accuracy (%)	81.55	65.42	−16.13
FGSM Robust Acc (%)	1.73	31.52	+29.79
PGD Robust Acc (%)	0.00	25.15	+25.15
Clean ECE	0.1278	0.1174	−0.0104
FGSM ECE	0.9477	0.4420	−0.5057
PGD ECE	1.0000	0.5399	−0.4601

TABLE II: ResNet-18: Standard vs. FGSM-AT.

The standard ResNet-18 achieves 81.55% clean accuracy — a significant improvement over the Small CNN — although it is more catastrophically vulnerable to adversarial attacks: robust accuracy drops to 1.73% under FGSM and exactly 0.00% under PGD. What is more, the adversarial ECE under PGD reaches 1.0, the theoretical maximum, meaning the model is 100% confident in every single wrong prediction. Larger capacity alone does not confer robustness.

After FGSM-AT, the ResNet-18 recovers to 31.52% FGSM accuracy and 25.15% PGD accuracy, a meaningful improvement though lower in absolute terms than the Small CNN. The robustness tax is more severe for ResNet-18: clean accuracy drops from 81.55% to 65.42%, a gap of ~ 16 pp vs. ~ 13 pp for the Small CNN. On clean data the two ResNet-18 variants start with nearly identical calibration ($ECE \approx 0.11$ – 0.13), mirroring the Small CNN pattern. Under attack, the standard model's ECE again goes up catastrophically, while the adversarial model keeps its ECE considerably lower across all ε values.



(a) Robust Acc. (b) Calibration (ECE) (c) Clean Trade-off

Fig. 2: ResNet-18 analysis.

IV. DISCUSSION

A. Does FGSM-AT improve robustness against FGSM and PGD?

Yes, consistently across both architectures. FGSM-AT improves both FGSM and PGD accuracy by a large margin compared to standard training. Critically, the improvement generalises to PGD — a strictly stronger attack — despite the model only seeing single-step examples during training. This “cross-attack transfer” is well-documented in the literature and arises because both attacks exploit the same gradient direction; multi-step PGD refines the same direction rather than finding a qualitatively different one.

B. Robustness vs. Calibration

Increased robustness correlates with worse calibration on clean data (higher ECE, higher NLL) and with substantially better calibration under attack. This apparent paradox resolves when the evaluation condition is fixed: adversarial training shifts probability mass away from high-confidence predictions, making the model slightly underconfident on clean samples but preventing the catastrophic overconfidence collapse observed in standard models under attack. For deployment in adversarial environments, the adversarially trained model’s calibration is therefore far more trustworthy.

C. Overconfidence vs. Underconfidence

Reliability diagrams reveal a consistent pattern: adversarially trained models tend to be slightly underconfident on clean data and moderately overconfident under attack. Standard models are well-calibrated on clean data but maximally overconfident under attack. The adversarial training objective, which maximises the loss on perturbed inputs, implicitly flattens the confidence distribution — a desirable property under distribution shift.

D. Effect of Model Scale

Scaling from the Small CNN to ResNet-18 amplifies both the benefits and the costs of adversarial training. The standard ResNet-18 achieves higher clean accuracy and better calibration on clean data, but it is paradoxically more fragile (PGD accuracy 0.00% vs. $\sim 0.3\%$). After adversarial training, the ResNet-18 recovers comparable absolute robustness to the Small CNN but pays a higher robustness tax in clean accuracy (~ 16 pp vs. ~ 13 pp). This suggests that larger models have more “clean accuracy capacity” to sacrifice, and that the interaction between model capacity and adversarial training is non-trivial: capacity does not substitute for adversarial training, but it does increase the cost of applying it.

V. CONCLUSION

This project confirms a clear robustness–calibration trade-off in deep neural networks trained on CIFAR-10. FGSM-based adversarial training substantially improves both FGSM and PGD robustness at the cost of reduced clean accuracy and a mild degradation in clean-data calibration. However, it dramatically improves calibration under attack, preventing the catastrophic overconfidence exhibited by standard models. Scaling to ResNet-18 reinforces these findings: larger capacity neither confers robustness automatically nor alleviates the robustness tax — it simply enlarges the magnitude of both effects. Future work could explore temperature scaling or other post-hoc calibration methods to recover clean-data calibration without sacrificing robustness.

TABLE III: Appendix: Summary of all experimental results.

Arch.	Train.	Clean	FGSM	PGD	ECE_{cl}	ECE_{fgsm}
SCNN	Std.	74.96	6.20	0.29	0.1591	0.8835
	FGSM-AT	62.00	35.25	29.13	0.1084	0.1152
RN18	Std.	81.55	1.73	0.00	0.1278	0.9477
	FGSM-AT	65.42	31.52	25.15	0.1174	0.4420